

PHY 412: Statistical Physics

25-26 Semester II

Assignment – 4

Problems in black are for submission; Blue ones are for practice. Due date: April 13th in Class

1. Express the correlation length ξ of an Ising chain in terms of the largest two eigen values λ_1 and λ_2 of the transfer matrix T in presence of a magnetic field.
2. We have a field-free Ising chain with nearest-neighbor and next-nearest-neighbor interactions, where the Hamiltonian is given by,

$$H(s_i) = -J_1 \sum_i s_i s_{i+1} - J_2 \sum_i s_i s_{i+2}$$

Solve this Ising model by introducing a new variable $\sigma_i = s_i s_{i+1}$ and derive the expressions for $\frac{1}{N\beta} \frac{\partial}{\partial J_1} Z$ and $\frac{1}{N\beta} \frac{\partial}{\partial J_2} Z$. What do they signify?

3. Consider a double Ising chain such that the nearest-neighbor coupling constant along either chain is J_1 while the one linking adjacent spins in the two chains is J_2 . Then, in the absence of the field,

$$H\{s_i, s'_i\} = -J_1 \sum_i (s_i s_{i+1} + s'_i s'_{i+1}) - J_2 \sum_i s_i s'_i.$$

Using the transfer matrix method, show that the partition function of this system is given by,

$$\frac{1}{2N} \ln Z \approx \frac{1}{2} \ln \left[2 \cosh K_2 \left\{ \cosh 2K_1 + \sqrt{(1 + \sinh^2 2K_1 \tanh^2 K_2)} \right\} \right],$$

where $K_1 = \beta J_1$ and $K_2 = \beta J_2$.

4. Write down the transfer matrix P for a one-dimensional spin-1 Ising model in zero field, described by the Hamiltonian $H\{s_i\} = -J \sum_{i=1}^{N-1} s_i s_{i+1}$ where $s_i = -1, 0, +1$. Show that the free energy of this model is given by, $-k_B T \ln \left[\frac{1}{2} \left\{ (1 + 2 \cosh \beta J) + \sqrt{8 + (2 \cosh \beta J - 1)^2} \right\} \right]$. Examine its limiting behavior at the limits $T \rightarrow 0$ and $T \rightarrow \infty$.
5. Consider the 1D Ising model $H\{s_i\} = -J \sum_i s_i s_{i+1} - h \sum_i s_i$. Show that the free energy is non-analytic at $T = 0$, but otherwise analytic. [Think about the behaviour of the magnetization, a derivative of the free energy.]
6. Consider the 1D field-free open-ended Ising model Hamiltonian: $H = - \sum_i J_i s_i s_{i+1}$.
 - (a) Show that the two-point correlation function, $C_i(r) \equiv \langle (s_i - \langle s_i \rangle)(s_{i+r} - \langle s_{i+r} \rangle) \rangle = \prod_{n=i}^{i+r-1} \tanh(\beta J_n)$.
 - (b) Show that the four-point correlation function, $\langle s_i s_j s_k s_l \rangle = \prod_{n=i}^{j-1} \tanh(\beta J_n) \prod_{n=k}^{l-1} \tanh(\beta J_n)$